

Swayam Patel

M.Tech – Software System

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Education

Dhirubhai Ambani University CPI: 8.40	2025 - Present Gandhinagar, Gujarat
Madhuben and Bhanubhai Patel Institute of technology CPI: 7.91	2021 - 2025 Vidya Nagar, Gujarat
Vidyamangal Residential School (GHSEB) Percentage: 70	2020 - 2021 Surat, Gujarat
Vidyamangal Residential School (GSEB) Percentage: 75	2017 - 2018 Surat, Gujarat

Experience

Dhirubhai Ambani University Teaching Assistant	Jun 2025 – Present Gandhinagar, Gujarat
- Assisted professors in delivering DBMS and ICT courses by supporting lectures, conducting lab sessions, and helping students with concepts, assignments, and practical work. - Evaluated assignments, resolved technical doubts, and provided academic guidance to enhance students understanding of database systems and information and communication technologies.	
Anand Asset Manager Pvt. Ltd. Python Developer Intern	Jan 2025 – May 2025 Anand, Gujarat
- Developed a high-performance system processing real-time tick data for 2000+ stocks with sub-second latency. - Implemented data aggregation pipelines with Redis Streams and Pandas to generate OHLC (Open, High, Low, Close) data at multiple time intervals. - Designed efficient data storage and retrieval mechanisms using PostgreSQL, ensuring scalability and low-latency access for market analytics. - Optimized system architecture for real-time processing, fault tolerance, and future scalability improvements.	

Projects

RAG-based Offline AI Assistant for PDF Querying RAG, Ollama, Streamlit	🔗
- Built a local AI system that answers questions from PDF documents using a Retrieval-Augmented Generation (RAG) pipeline . - Generated high-quality document embeddings using Sentence-Transformers and implemented semantic search with FAISS for efficient retrieval. - Integrated a locally hosted LLaMA model via Ollama to ensure complete data privacy without cloud dependency. - Developed an interactive chat interface using Streamlit for real-time question answering. - Optimized retrieval pipeline for faster response time and improved contextual accuracy.	
AI-Powered Plant Disease Detection System with LLM Assistant Keras, MobileNetV2, Gemini API	🔗
- Developed an AI-based web application to detect plant diseases from leaf images and provide real-time disease prediction with confidence scores. - Trained and fine-tuned a MobileNetV2 transfer learning model on the PlantVillage dataset containing 39 plant disease classes and 60K+ images, achieving around 85–92% accuracy. - Built an end-to-end Flask backend for image upload, preprocessing, normalization, and model inference. - Integrated an LLM assistant using Gemini to generate farmer-friendly explanations including symptoms, causes, prevention methods, and treatment suggestions.	

ResearchMind — Multi-Agent AI Research System | Python, Streamlit, LangChain, Gemini API, Tavily API, BeautifulSoup

- Developed a multi-agent AI research assistant that takes a user topic and automatically generates a structured research report using a 4-stage pipeline: search, scrape, write, and critique.
- Implemented specialized agents including a Search Agent using Tavily API, Reader Agent for web scraping, Writer Chain for report generation, and Critic Chain for evaluation and scoring.
- Optimized the system to use only 2 LLM API calls per query, reducing API cost and making the application suitable for free-tier usage.
- Built an interactive Streamlit UI where users can enter research topics, view generated reports, receive critic feedback, and download reports as Markdown files.

Technical Skills

Languages: C++, Python, SQL
Developer Tools: Git, Docker, VS Code, Figma
Frameworks: Node.js, Tailwind CSS, Bootstrap
Cloud/Databases: MongoDB, MySQL, AWS S3
Soft Skills: Team Collaboration, Leadership, Problem Solving
Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning
Areas of Interest: Generative AI, System Design, Competitive Programming, AI/ML, Cloud Computing

Position Of Responsibility

Teaching Assistant <i>Dhirubhai Ambani University</i>	Jun 2025 - Present
• Assisted the professor in managing lectures, preparing course materials, and supporting students during classes and lab sessions.	

Achievements

- Qualified GATE 2025 in Computer Science with a score of 496 and an All India Rank (AIR) of 7171.
- Qualified GATE 2025 in Data Science and Artificial Intelligence (DA) with a score of 346 and an All India Rank (AIR) of 8603.
- **Supervised Machine Learning – Coursera (2025):** Completed a certified Machine Learning course, gaining hands-on experience in algorithms, model training, and data analysis.